Andrew Boldi

Solving hard science problems with machine learning

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Education

UC Berkeley — *BS in Chemical Biology & BA in Computer Science* *Expected May 2027*

Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Computer Architecture, Deep Neural Networks & Computer Vision (Fall '26), Computational Chemistry, Physical Chemistry (Quantum & Statistical Mechanics), Honors Organic Chemistry I/II, Chemical Biology (Biochemistry)

Publications

A. Boldi†, A. K. Gupta†, Y. Tang, M. Hamadani, K. Ameri, Y. Yeghiazarians. "Fragment-Centric Explainability of Chemical Foundation Models with Attention-Aware Layer-Wise Relevance Propagation." *Under review*, 2026. †Co-first authors.

Work Experience

AI Intern — *Soley Therapeutics: Molecular Machine Learning (Dr. Ankur Gupta)* *South San Francisco, CA*
May 2025 – Present

- Full-time intern (summers 2025 & 2026); received return offer and continued part-time during the academic year
- Developed fragment-centric explainability framework for chemical foundation models (**co-first author**; patent filed)
- Leading computer vision explainability for cell images; developing generative chemistry models
- Finetuned hundreds of models on public and proprietary data to predict molecular target candidate profiles (TCPs); built automated benchmarking framework spanning 22 datasets (TDC ADMET group)
- Implemented and benchmarked several architecture ideas for molecular property prediction foundation models
- Automated CDD protocol creation (Selenium + Python), completing a 100+ manual-hour task in 5–6 hours
- Tools used: PyTorch, PyTorch Lightning, Hydra/OmegaConf, CDD Vault, AWS (EC2, S3, CodeBuild)

Research Experience

Summer Research Intern — *Caltech: Stoltz Lab Group (Dr. Scott Virgil)* *Pasadena, CA*
Summers 2022 & 2023

- Completed 16 consecutive steps toward scabrolide B; synthesized korormicin & ineleganolide fragments
- Performed computational modeling (ORCA) to determine selectivity in a late-stage stereoinversion
- Developed StremLigand, a catalyst & ligand cataloging application

Assistant Researcher — *Caltech: Center for Catalysis and Chemical Synthesis* *Remote*
Jul 2020 – May 2022

- Developed remote-automation software for robotic autosamplers (Python, VBA, M68k assembly); used it to screen lanthanide-ligand catalysts for an asymmetric reduction

Projects

Molecular Optical Rotation Prediction with Machine Learning — *Trennen* *2022*

- Predicted rotation direction of optically active molecules with a T-Net (63% accuracy, 50% random baseline); distilled 8M+ molecules to ∼27,000 optically active ones via PubChem API, RDKit, and Open Babel

Scenic Navigation Algorithm with Deep Learning — *Pathfinder* *2022*

- Built scenic routing algorithm (modified A*); segmented 250K+ scraped Street View images on Google TPUs

Pacman with Bitcoin Miner — *Video Demo* *Spring 2024*

- Top-15 project among ∼1400 students in UC Berkeley's data structures course (CS 61B); 3000+ lines of Java

Math Competition Training Website — *AMC Trainer* *2021 – Present*

- Built training site for math competitions, used by 20K+ users in 50+ countries (Cloudflare Workers, Node.js)

Skills

Programming: Python, Java, Rust, C/C++, JS/CSS/HTML, Svelte, Swift, M68k Assembly, LaTeX
ML & Software: PyTorch, PyTorch Lightning, TensorFlow, RDKit, Hydra/OmegaConf, Selenium, AWS, Git, Slurm, Linux
Chemistry: Multistep organic synthesis, HPLC, LC/MS, NMR, computational modeling (ORCA, DFT)
Languages: English, Armenian, Spanish

Additional

- **Y Combinator AI Startup School (2025):** one of 2000 selected attendees (∼20 invited to partner dinner)
- **Piano:** performed at Carnegie Hall and Merkin Hall; 30+ competition recognitions (YouTube)
- **Rubik's cubes:** solved 3x3 in 9.42 seconds, including blindfolded solves; 2x2 through 7x7, Megaminx, Square-1